DS605 - Fundamentals of Machine Learning

Lecture 04 - Measures of Central Tendency & Dispersion

Lecture Notes

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Overview

These notes accompany the Lecture 04 slide deck. Lectures 02-03 introduced ShopCart customers as vectors and compared them using distance and angle. This lecture first asks a simpler, more fundamental question: given a thousand ShopCart customers, what single number best represents the whole group? The answer depends entirely on which notion of "average" is chosen, and the wrong choice can quietly mislead every decision built on top of it. The lecture is in two parts: Part 1 covers measures of central tendency (mean, median, mode - "what is typical?"), and Part 2 covers measures of dispersion (range, quantiles, variance, standard deviation - "how spread out is everything else?"). A number that summarizes the center of a dataset is only as trustworthy as the spread around it, so the two halves of this lecture are really one idea told in two parts.

LEARNING OBJECTIVES

By the end of this lecture you should be able to: (1) compute the mean, median and mode of a dataset and know when each is the right choice; (2) explain why the mean is sensitive to outliers while the median and mode are not; (3) compute range, quantiles, the interquartile range, variance, standard deviation and the coefficient of variation; (4) read a boxplot or violin plot and relate it to the five-number summary; (5) apply the 68-95-99.7 rule; and (6) translate every one of these formulas into a single line of Python (NumPy / SciPy).

Part 1 - Measures of Central Tendency

A measure of central tendency is a single value meant to represent an entire distribution - informally, "where would most of the values fall if you had to guess with one number?" There are three standard candidates - the mean, the median, and the mode - and, crucially, they are not interchangeable. A quick way to see this is a thought experiment: if HR asked everyone in a room, "What is the average salary at ShopCart?", different people would report different numbers depending on which "average" they had in mind, and none of them would be wrong. That disagreement is the entire subject of Part 1.

1.1 The Mean

The mean (arithmetic average) is the measure everyone learns first. Let x1, x2, ..., xN be the N observed values of an attribute X. Each value may optionally carry a weight w_i, giving the weighted mean:

$$\bar{x} = \frac{1}{N}\sum_{i=1}^{N} x_i \qquad \bar{x}_w = \frac{\sum_{i=1}^{N} w_i\, x_i}{\sum_{i=1}^{N} w_i}$$

ShopCart interpretation: the mean order value across all of today's transactions.

1.2 The mean is highly sensitive to outliers

Suppose 50 ShopCart employees have salaries between Rs 25,000 and Rs 60,000; the mean is a fair summary at roughly Rs 40,000. Now, suppose the founder's row is accidentally included in the same spreadsheet, with a salary of Rs 5,00,00,000. The mean rockets past Rs 1,00,000 - more than double - even though 50 of the 51 people still earn exactly what they always earned. One extreme value dragged the "typical" number far away from what is actually typical.

> **PARTIAL FIX - THE TRIMMED MEAN**
>
> Chop off the extreme values (for example, the top and bottom 5%) before averaging. This reduces sensitivity to outliers, but trimming too aggressively risks Loss of Information (LoI): real signal, not just noise, gets discarded along with the extremes.

1.3 The Median

The median is the middle value once the data is sorted, and it is robust to outliers because it only cares about rank, not magnitude.

- Sort the values from smallest to largest.
- Odd N: take the middle value.
- Even N, numeric attribute: average the two middlemost values.
- Even N, ordinal attribute: any value between the two middlemost values is a valid median (there is no "average" of two categories).

$$\text{Median} = x_{(\frac{N+1}{2})} \quad (N \text{ odd})$$

Economists favor the median for exactly the reason the founder-salary example above illustrates: CEO salaries, house prices, startup valuations, and national income all contain a handful of extreme values that would drag a mean far from what a typical household or company actually experiences. For skewed data, the median is the better measure of central tendency.

1.4 The Mode

The mode is the most frequently occurring value. It is the only one of the three measures that is defined for qualitative (categorical) attributes as well as quantitative ones - you cannot take the mean of a set of T-shirt sizes, but you can ask which size is bought most often. A distribution can have more than one mode; it is then called multimodal (bimodal, trimodal, and so on).

> **BIMODAL EXAMPLE - CONTROVERSIAL QUESTIONS POLARIZE**
>
> Survey responses to "How do you feel about capital punishment?": strongly oppose 42, somewhat oppose 6, neither 3, somewhat support 4, strongly support 45. Two peaks sit at the extremes with almost nobody in the middle - a single "average" opinion would misrepresent the room entirely.

1.5 Choosing the right measure

The right measure depends on both the attribute type and the distribution's shape. Not every measure is even defined for every attribute type - a nominal attribute like city has no natural ordering, so mean and median are meaningless for it; only the mode is defined. The table below, worked through in class, summarizes the decision.

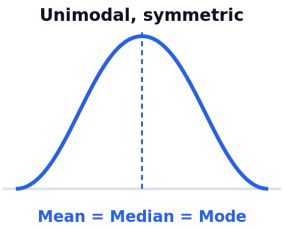
Attribute type	Measure(s) defined	Best measure	Why
Nominal	Mode only	Mode	No ordering, so the mean and median are undefined

Attribute type	Measure(s) defined	Best measure	Why
Ordinal	Mode, Median	Median	Ordering exists, but the gaps between categories are not equal
Numeric (symmetric)	Mean, Median, Mode	Mean	Uses every value, and coincides with the median and mode anyway
Numeric (skewed)	Mean, Median, Mode	Median	Robust to the outliers that create the skew

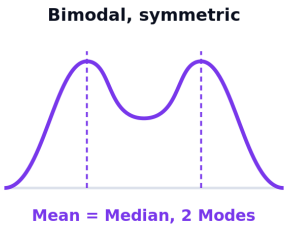
Three quick checks worth internalizing: not every dataset has all three measures defined (a purely nominal dataset has no mean); in a normal (symmetric, bell-shaped) distribution the mean, median and mode coincide; and among the three, the mean and median are always single, unique values, while the mode may be non-unique (multimodal) or, for continuous data with no repeated values, may not exist in a useful sense.

1.6 Symmetric versus skewed distributions

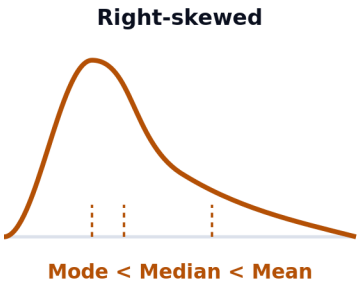
When a distribution is symmetric about its center (skewness = 0), the mean, median, and mode coincide, and any of the three will do. Symmetrical, bell-shaped data is, however, the exception rather than the rule: house prices, personal income, social-media follower counts, daily stock returns, network traffic, and ShopCart order values are all skewed in practice. When a distribution is skewed, a long tail on one side pulls the mean toward that tail, while the median stays anchored near where most of the data actually sits - which is why the ordering of mean, median and mode is a quick visual diagnostic for the direction of skew (mode less than median less than mean indicates a right/positive skew; the reverse indicates a left/negative skew).



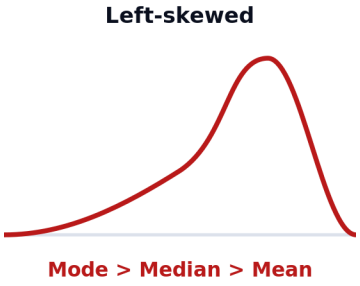
Symmetric data: mean, median, and mode coincide (unimodal and bimodal cases).



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Skewed data: the three measures are separate, with the mean pulled furthest toward the long tail.



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1.7 Central tendency, in code

Every formula above is exactly one line of Python once the data is in a NumPy array or pandas Series.

```
import numpy as np
from scipy import stats

order_values = [420, 680, 150, 990, 420, 730, 310, 420, 860, 275]

np.mean(order_values)                     # mean
np.median(order_values)                   # median
stats.mode(order_values, keepdims=False)  # mode
```

SECTION SUMMARY

Choosing the right "average" is not a statistics footnote - it is the number that ends up on the dashboard everyone acts on. Use the mean when the data is roughly symmetric and every value should count; use the median when outliers or skew are present; use the mode for categorical data or to describe the most common case.

Part 2 - Measures of Spread (Dispersion)

A single measure of central tendency is never the whole story. A measure of dispersion (or spread) describes how variable a sample or population is - how far, on average, individual values wander from the center. Spread tells us how well the central-tendency number actually represents the data: a large spread means the "average" is a weaker summary of any individual observation.

2.1 Motivating example - same mean, different reality

Store A and Store B both average exactly 80 orders per day. Are they the same business to run?

Day	Store A	Store B
Mon	80	20
Tue	81	80
Wed	79	100

Day	Store A	Store B
Thu	80	120

Store A hovers tightly around 80 every day; Store B swings from 20 to 120. Staffing, inventory and cash-flow planning for Store B is a fundamentally harder problem, even though its mean is identical to Store A's. This is the entire motivation for the rest of Part 2: a number that captures exactly what the mean cannot.

2.2 Range

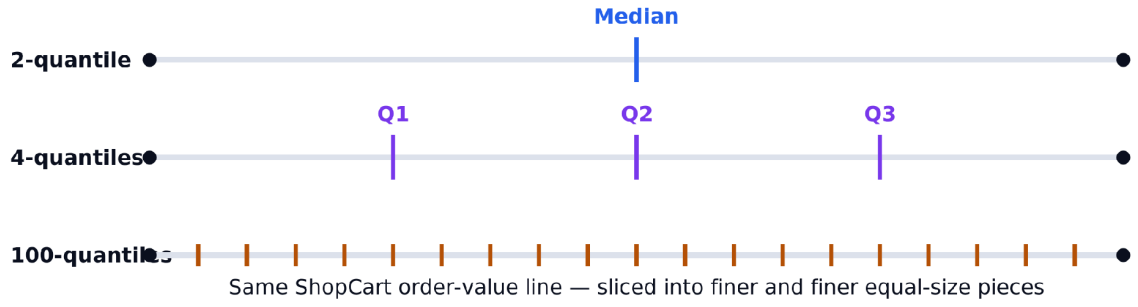
The simplest spread measure is the distance between the largest and smallest observed values.

$$\text{Range} = \max(X) - \min(X)$$

Store B's range is 120 minus 20, which equals 100; Store A's range is 81 minus 79, which equals 2. One number already tells much of the story, but the range only looks at the two most extreme points and ignores everything in between - it is highly sensitive to a single outlier and blind to how the rest of the data is distributed.

2.3 Quantiles

Quantiles are data points that split a distribution into equal-sized consecutive sets. The 2-quantile (splitting the data into a lower and an upper half) is exactly the median; the 4-quantiles are the quartiles (three points splitting the data into four equal parts); the 100-quantiles are the percentiles (ninety-nine points splitting the data into a hundred equal parts).



The same ShopCart order-value line, sliced into finer and finer equal-size pieces: 2, 4, and 100 quantiles.

Formally, the k-th q-quantile is the value x such that at most k/q of the data lies below x and at most (q minus k)/q lies above it, for an integer k with 0 less than k less than q. There are exactly q minus 1 such q-quantiles for any given q.

$$x \text{ such that at most } \frac{k}{q} \text{ of the data} < x \text{ and at most } \frac{q-k}{q} > x$$

2.4 Interquartile Range (IQR)

The IQR is the distance between the first and third quartiles - the range covered by the middle half of the data.

$$IQR = Q_3 - Q_1$$

A common rule of thumb flags any value more than 1.5 times the IQR above Q3 or below Q1 as a suspected outlier. Why 1.5 and not 1 or 3? For normally distributed data, 1.5 times IQR corresponds to roughly plus or minus 2.7 standard deviations from the mean, which flags only about 0.7% of points as outliers - a threshold tight enough to be useful but loose enough to avoid flagging ordinary variation. A stricter multiplier, like 1, would flag far too many ordinary points; a looser one, like 3, would miss real outliers.

2.5 Why the median and IQR resist outliers

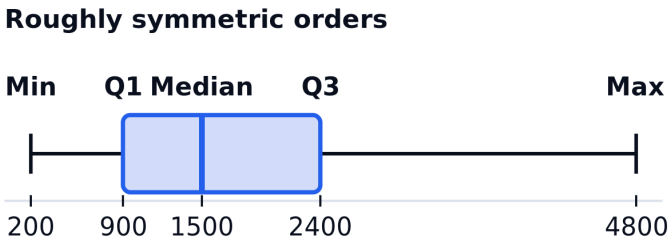
Consider 100 typical ShopCart retail orders (Rs 500 to Rs 3,000) plus one enterprise "whale" order of Rs 50,00,000.

Statistic	Before the whale order	After the whale order
Mean	Rs 1,540	Rs 51,020 (a 33x jump)
Median	Rs 1,510	Rs 1,515 (barely moves)
IQR	Rs 980	Rs 1,010 (barely moves)

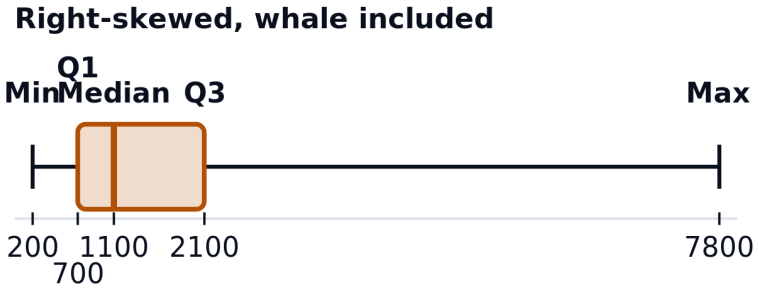
The mean is not just sensitive to outliers - a single data point can single-handedly hijack it. The median and IQR both shrug off the whale because they only care about rank, not magnitude.

2.6 The Five-Number Summary, boxplots, and violin plots

Median and quartiles alone say nothing about the tails, so the five-number summary adds the two extremes: minimum, Q1, median, Q3, maximum. A boxplot is the standard visualization of this summary: the box spans from Q1 to Q3 (so the box length is the IQR), the median is drawn as a line inside the box, and whiskers extend out to the minimum and maximum observations (or to 1.5 times IQR with individual outlier points marked beyond that, in the more common convention).



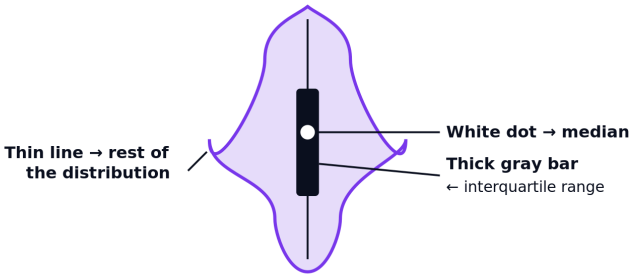
Boxplots of a roughly symmetric order-value distribution and a right-skewed one with a long whisker toward the whale customer.



Long right whisker = enterprise orders pulling the tail out.

Boxplots of a roughly symmetric order-value distribution and a right-skewed one with a long whisker toward the whale customer.

A violin plot is a richer alternative: it shows the same summary statistics together with the density (shape) of the distribution - effectively a boxplot with the actual silhouette of the data drawn around it, revealing multimodality that a plain boxplot would hide.



A violin plot: the thick gray bar is the interquartile range, the white dot is the median, and the thin line and silhouette show the rest of the distribution.

Between two boxplots, the visual cue that instantly signals skew is asymmetry: a longer whisker or box on one side, and a median line sitting off-center within the box, rather than in the middle.

2.7 Variance and Standard Deviation

Quantiles ignore most individual scores - a boxplot is built from just five numbers. To get a fully representative measure of spread, every value in the dataset must be used, not just a few ranked points. That is exactly what variance does: it averages the squared distance of every observation from the mean.

$$\sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(x_i - \bar{x})^2 \qquad \sigma = \sqrt{\sigma^2}$$

Returning to Store A vs. Store B:

	Values	Mean	Variance	Std. dev.
Store A	80, 81, 79, 80	80	0.5	0.71
Store B	20, 80, 100, 120	80	1,300	36.1

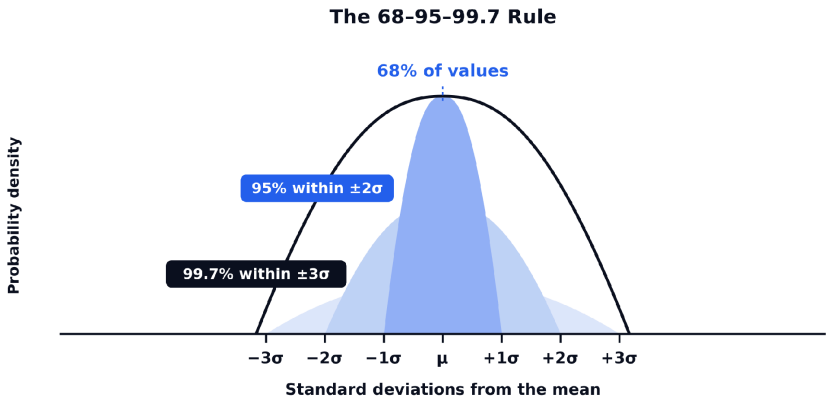
Same mean, 2,600 times the variance. This is the number that was missing from the motivating example at the start of Part 2 - it is now obvious why Store B is the harder business to run, even though its average daily order count looked identical to Store A's.

2.8 Properties of standard deviation

- Standard deviation measures spread about the mean, so it is only meaningful when the mean is the chosen measure of central tendency (pairing it with the median would be inconsistent).
- Standard deviation equals zero only when there is no spread at all - every observation is identical. Otherwise, it is strictly positive.
- A low standard deviation means observations cluster tightly around the mean; a high one means observations are spread across a wide range.

2.9 The 68-95-99.7 Rule

For roughly normally distributed data, approximately 68% of values fall within one standard deviation of the mean, approximately 95% fall within two, and approximately 99.7% fall within three. This single rule underlies z-score anomaly detection, confidence intervals, and statistical process control charts, and it will resurface repeatedly in machine learning.



The 68-95-99.7 rule: the proportion of a normal distribution captured within 1, 2, and 3 standard deviations of the mean.

2.10 Coefficient of Variation (CV)

Standard deviation is expressed in the same units as the data, so it cannot be used to directly compare the "spread" of two attributes measured on different scales. The coefficient of variation normalizes standard deviation by the mean, producing a unitless ratio that is comparable across attributes:

$$CV = \frac{\sigma}{\mu}$$

WORKED EXAMPLE

ShopCart order value: mean Rs 2,000, std dev Rs 600, giving CV = 0.30. Session duration: mean 8 minutes, std dev 3 minutes, giving CV = 0.375. Comparing raw standard deviations (Rs 600 vs. 3 minutes) is meaningless because the units differ; the CV reveals that session duration is the more volatile of the two metrics.

2.11 Dispersion, in code

```
import numpy as np
from scipy import stats

order_values = [420, 680, 150, 990, 420, 730, 310, 420, 860, 275]

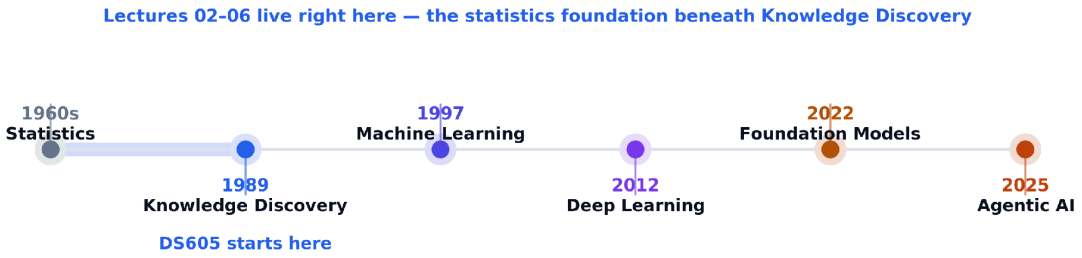
np.var(order_values)                    # variance
np.std(order_values)                    # standard deviation
np.percentile(order_values, [25, 75])   # Q1, Q3 -> IQR
stats.variation(order_values)           # coefficient of variation
```

SECTION SUMMARY

Spread is often the more important number of the two halves of this lecture - it is what tells you whether the average can be trusted. A dashboard that reports only a mean is incomplete; pairing it with a standard deviation, an IQR, or a full boxplot turns a single number into an honest picture of the data.

Where this leads

Lectures 02-06 all build the statistical foundation beneath Knowledge Discovery: mean, variance, and distance are the load-bearing walls that everything from Lecture 07 onward is built on. Every technique that resurfaces later - anomaly detection, clustering, normalization, confidence intervals, feature scaling - starts by asking exactly the question this lecture asked: what is typical, and how spread out is everything else?



The map of Lectures 02-06 within the broader Knowledge Discovery timeline.

Lecture 04 - the whole toolkit

No heavy coding is required for this course - just enough Python to connect every formula in this lecture to a real, runnable line of code.

```
import pandas as pd
from scipy import stats

df = pd.DataFrame({"order_value": [420, 680, 150, 990, 420, 730, 310, 420, 860, 275]})
x = df["order_value"]

x.mean(), x.median(), stats.mode(x, keepdims=False)
x.std(), x.var(), stats.variation(x)
q1, q3 = x.quantile([0.25, 0.75])
iqr = q3 - q1
df.boxplot(column="order_value")
```

Quick reference - Lecture 04

Concept	Formula	Python
Mean	mean = (1/N) sum of x_i	np.mean(x)
Median	middle value once sorted	np.median(x)
Mode	most frequent value	scipy.stats.mode(x)
Range	max(X) - min(X)	np.ptp(x)
Quantile/percentile	value splitting data into equal parts	np.percentile(x, q)
IQR	Q3 - Q1	np.percentile(x,75) - np.percentile(x,25)
Variance	average squared deviation from the mean	np.var(x)
Standard deviation	square root of variance	np.std(x)
Coefficient of variation	std dev / mean	scipy.stats.variation(x)